Four models, every completed learning stage

MobileCLIP2-S2 • SigLIP2-B/16-224 • SmolVLM-500M-Instruct • VadCLIP

What is compared

Zero-shot / original model
Frozen-feature linear probes
Partial visual fine-tuning
Language LoRA and temporal adapters
Later region / source-data experiments

How to read the results

Five target-specific binary tasks
Dataset and positive counts shown
Ranking + fixed-cutoff metrics
Primary development-selected routes
Unavailable stages explicitly marked

This deck covers training and accuracy evaluation; all numbers come from preserved runs.

The stages differ by model architecture

A stage is reported only when a corresponding saved experiment exists

Model	Baseline	Linear probe	Weight adaptation	Later stages
MobileCLIP	Text margin	Yes	Last visual stage	Crops / GRU / source fits
SigLIP2	Text margin	Yes	Last visual layers	Geometry / ensembles
SmolVLM	Original binary query	Not run	Language-layer LoRA	Joint / routed / transfer
VadCLIP	CLIP-B/16 text	Yes	Temporal adapter	Branch selection

> **No study unfreezes every parameter: visual tuning, LoRA and adapter training have distinct meanings.**

Smoking: all four model families, stage by stage

AP overview • A and B are different tests • the adaptation type differs by architecture

Model	Test N / positive	Zero-shot / original	Linear probe	Weight-adapted
MobileCLIP	A: 646 / 393	0.642	0.594	0.564
SigLIP2	A: 646 / 393	0.669	0.670	0.650
SmolVLM	A: 646 / 393	0.600	Not run	0.623
VadCLIP	B: 1,056 / 560	0.482	0.544	0.633

A: camera-c1 holdout. B: session holdout across cameras. These rows do not establish a four-model winner.

Overall data coverage, without double counting studies

Rows are study manifests and reuse sources • counts are examples, not independent events

Study	Train	Dev	Test / transfer	Total / unit
CLIP visual study	3,258	1,633	1,801 + 806 external	7,498 mixed examples
Strategy / Smol main	3,973	1,812	2,025 + 806 external	8,616 mixed examples
Additional transfer	0	0	906 + 28 + 27	961 clips/photos/windows
VadCLIP smoking	1,910	312	1,056	3,278 original frames
HMDB adaptation	530	161	205	896 clips after purge
20 Oct XRF55 proxy	340	160	240	740 scripted clips

The 8,616-example corpus contains the earlier 7,498; these totals must not be added as unique data.

What the five targets are actually trained on

Input units and annotation scope determine what each evaluation supports

Objective	Sources used	Main evaluation unit / limitation
Smoking	CCTV; Stanford40; Kaggle; HMDB; XRF55	Frames/photos/actions; XRF55 is a gesture proxy
Littering	MIVIA-IWDD	Whole waste-dumping clips; broader than small littering
Loitering	Street Scene track/type labels	20-second windows; one camera, supervised re-split
Public sleep	Commons photos and matched controls	Apparent visible sleep; provisional review
Unattended	CAVIAR / ABODA; MEVA transfer	Overlapping endpoints; ownership/state labels limited

> **Primary-action labels, person boxes and posture labels are not interchangeable with verified events.**

Primary CLIP dataset: total / positive by target

7,498 target-specific examples • 12,478 distinct view images • one known objective per row

Target	Train N / +	Dev N / +	Test N / +	External N / +
Smoking (all sources)	2,097 / 716	1,135 / 413	646 / 393	806 / 140
Littering / IWDD	274 / 144	61 / 32	100 / 50	—
Loitering	323 / 41	98 / 13	75 / 11	—
Public sleep	74 / 25	21 / 7	18 / 6	—
Unattended	490 / 74	318 / 181	962 / 242	—

> **Only the annotated objective contributes loss; the other four labels remain unknown.**

Expanded strategy and SmolVLM dataset: total / positive

8,616 examples • 13,596 distinct decoded images • same expanded train/dev splits

Target	Train N / +	Dev N / +	Test N / +	External N / +
Smoking (all sources)	2,812 / 1,073	1,314 / 502	870 / 505	806 / 140
Littering / IWDD	274 / 144	61 / 32	100 / 50	—
Loitering	323 / 41	98 / 13	75 / 11	—
Public sleep	74 / 25	21 / 7	18 / 6	—
Unattended	490 / 74	318 / 181	962 / 242	—

The main expansion is smoking photos: +715 train, +179 dev and +224 held-out photos.

VadCLIP uses a separate session-held-out smoking split

Frozen CLIP-B/16 features • original 3,278 annotated frames across 16 camera/date sessions

Split	Sessions	Frames	Smoking +	Normal −	Prevalence
train	9	1910	756	1154	39.6%
dev	4	312	106	206	34.0%
test	3	1056	560	496	53.0%

Train and test share physical cameras across dates; this is not the CLIP camera-c1 holdout.

Transfer cohorts answer different generalization questions

These are separately reported cohorts; their labels and train/test roles are explicit

Cohort	Held-out N / +	Role	What is still limited
Stanford40	806 / 140	Official photo test	Photo domain contributes training
HMDB action transfer	906 / 109	Unseen-source test in strategy/Smol	Other actions are non-exhaustive negatives
New-author public sleep	28 / 17	No training or selection	25 author groups; provisional labels
MEVA abandonment	27 / 15	No training or selection	Action windows, not persistent item state
XRF55 scene 4	240 / 60	20 Oct new-source proxy test	One scene, three subject aliases; masked gestures

> **The later HMDB 205-clip adaptation test and the earlier 906-clip transfer test must stay separate.**

Train, select and test play different roles

Development data determine checkpoints, routes and cutoffs; test data report the frozen result

Training and validation

Task / class / source balancing
Grouped camera / recording / author splits
Unknown targets have zero loss
File and decoded-pixel checks
Primary seeds specified in advance

Held-out reporting

All stage results remain visible
No best-test-seed substitution
No test-fitted decision threshold
Historical results labelled retrospective
Per-source denominators retained

Seeds estimate optimization variation; they do not add cameras or independent event evidence.

Use ranking and decision metrics together

All metrics are binary and source-specific • accuracy is exported with balanced accuracy

Metric / point	Meaning	How it is used here
AP / PR area	Precision–recall ranking across score cutoffs	Higher is better; compare the same labelled cohort
ROC-AUC	Pairwise positive/negative ranking	Higher is better; show with AP and prevalence
Precision / recall / F1	TP/(TP+FP); TP/P; harmonic mean	Use a named, frozen cutoff; report confusion counts
FPR / accuracy	FP/N; (TP+TN)/all	FPR matters for normal scenes; accuracy can hide imbalance
Raw / dev-max-F1	Zero margin / best development F1	No test recalibration; main stage tables use dev-max-F1
Dev FPR budget	At most 5% empirical development FPR	Test FPR can exceed 5%; not an alert-per-hour rate

AP can rise while FPR becomes unacceptable. A classifier score is not an event probability.

Linear probing and visual tuning change different weights

Same primary dataset and development-selection rule • five binary objectives

Setting	Frozen-feature probe	Visual fine-tune
Trainable part	Five logistic-regression classifiers	Last visual stage/layers plus five heads
Representation	Normalized last / mean / max / change	Same pooled representation; learned visual features
Optimization	C = 1, 10, 100; at most 2,000 LBFGS steps	AdamW; visual LR 1e-5 or 3e-5; head LR 3e-4
Regularization	Task/class/source/group sample weights	Weight decay .01; dropout .1; clipping 1.0
Preserve pretraining	Entire visual encoder frozen	Feature MSE ×.1; parameter anchor ×.1
Budget / inputs	C selected on dev macro source AP	Batch 32; max 10 epochs; patience 3; BF16 train

Earlier visual layers, text encoders and batch-normalization statistics remain frozen.

Six actual visual fine-tuning runs were completed

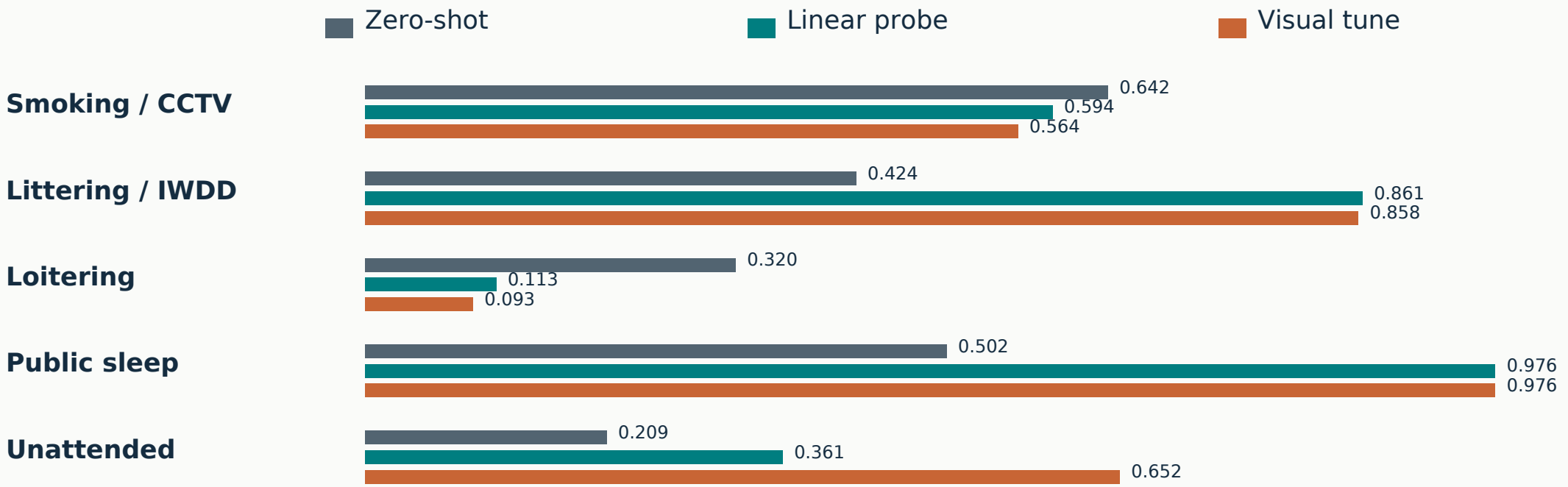
Primary seed 42; seeds 43 and 44 repeat the selected visual learning rate

Encoder	Visual params	Head params	Probe C	Visual LR	Best epoch
MobileCLIP	18,106,992	10,245	10.0	3e-05	3
SigLIP2	14,176,512	15,365	100.0	3e-05	2

Primary MobileCLIP chooses epoch 3; SigLIP chooses epoch 2. Later epochs do not replace them on test.

MobileCLIP: every training stage on the same holdout

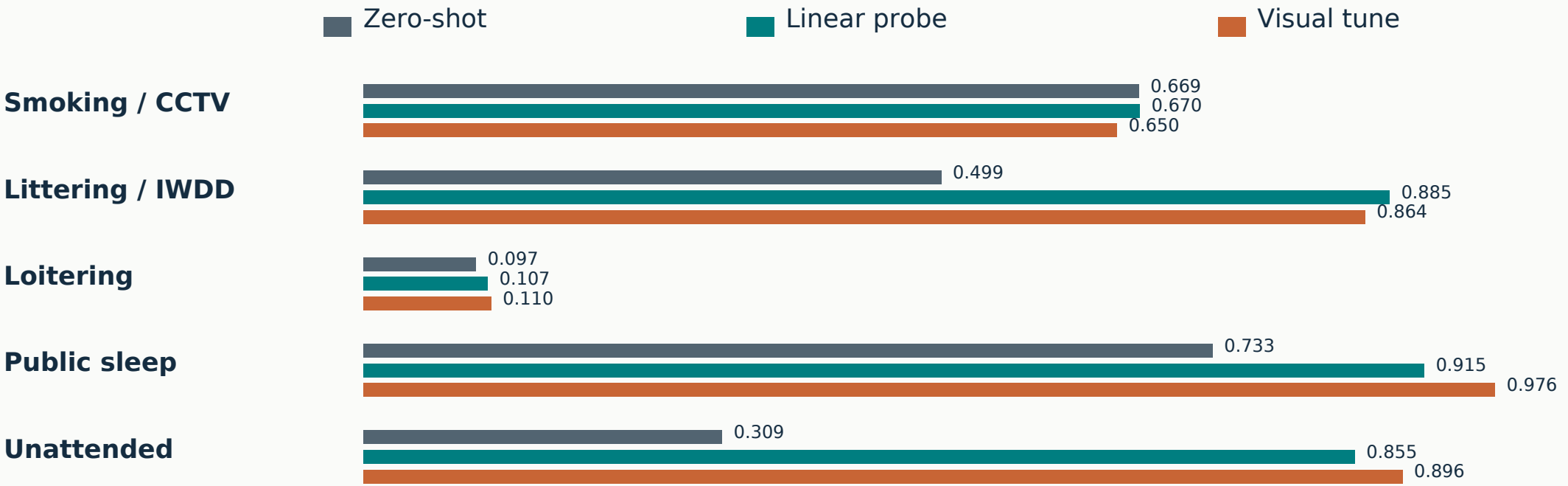
15–16 Oct study • primary seed 42 • AP by target; selected routes are reported separately



Training improves several rankings; smoking transfer and loitering remain inconsistent.

SigLIP2: every training stage on the same holdout

15–16 Oct study • primary seed 42 • AP by target; selected routes are reported separately



Training improves several rankings; smoking transfer and loitering remain inconsistent.

Smoking / CCTV: zero-shot → probe → visual tune

Same held-out cohort: n=646; positives=393; negatives=253 • dev-max-F1 cutoffs

Model / source	Stage	AP ↑	ROC-AUC ↑	F1 ↑	Prec. ↑	Recall ↑	FPR ↓
MobileCLIP	Zero-shot	0.642	0.543	0.749	60.6%	98.0%	98.8%
MobileCLIP	Linear probe	0.594	0.499	0.752	61.0%	98.0%	97.2%
MobileCLIP	Visual tune	0.564	0.440	0.735	60.7%	93.1%	93.7%
SigLIP2	Zero-shot	0.669	0.599	0.529	66.9%	43.8%	33.6%
SigLIP2	Linear probe	0.670	0.575	0.706	65.2%	76.8%	63.6%
SigLIP2	Visual tune	0.650	0.531	0.727	60.3%	91.3%	93.3%

All stages are shown, including regressions. These are candidate results, not test-selected routes.

Littering / IWDD: zero-shot → probe → visual tune

Same held-out cohort: n=100; positives=50; negatives=50 • dev-max-F1 cutoffs

Model / source	Stage	AP ↑	ROC-AUC ↑	F1 ↑	Prec. ↑	Recall ↑	FPR ↓
MobileCLIP	Zero-shot	0.424	0.347	0.611	46.8%	88.0%	100.0%
MobileCLIP	Linear probe	0.861	0.890	0.847	77.0%	94.0%	28.0%
MobileCLIP	Visual tune	0.858	0.872	0.841	78.9%	90.0%	24.0%
SigLIP2	Zero-shot	0.499	0.520	0.667	50.0%	100.0%	100.0%
SigLIP2	Linear probe	0.885	0.881	0.811	76.8%	86.0%	26.0%
SigLIP2	Visual tune	0.864	0.868	0.789	70.3%	90.0%	38.0%

All stages are shown, including regressions. These are candidate results, not test-selected routes.

Loitering: zero-shot → probe → visual tune

Same held-out cohort: n=75; positives=11; negatives=64 • dev-max-F1 cutoffs

Model / source	Stage	AP ↑	ROC-AUC ↑	F1 ↑	Prec. ↑	Recall ↑	FPR ↓
MobileCLIP	Zero-shot	0.320	0.814	0.468	30.6%	100.0%	39.1%
MobileCLIP	Linear probe	0.113	0.321	0.272	15.7%	100.0%	92.2%
MobileCLIP	Visual tune	0.093	0.151	0.000	0.0%	0.0%	12.5%
SigLIP2	Zero-shot	0.097	0.175	0.035	2.2%	9.1%	70.3%
SigLIP2	Linear probe	0.107	0.283	0.256	14.7%	100.0%	100.0%
SigLIP2	Visual tune	0.110	0.305	0.256	14.7%	100.0%	100.0%

All stages are shown, including regressions. These are candidate results, not test-selected routes.

Public sleep: zero-shot → probe → visual tune

Same held-out cohort: n=18; positives=6; negatives=12 • dev-max-F1 cutoffs

Model / source	Stage	AP ↑	ROC-AUC ↑	F1 ↑	Prec. ↑	Recall ↑	FPR ↓
MobileCLIP	Zero-shot	0.502	0.556	0.375	30.0%	50.0%	58.3%
MobileCLIP	Linear probe	0.976	0.986	0.833	83.3%	83.3%	8.3%
MobileCLIP	Visual tune	0.976	0.986	0.909	100.0%	83.3%	0.0%
SigLIP2	Zero-shot	0.733	0.764	0.600	75.0%	50.0%	8.3%
SigLIP2	Linear probe	0.915	0.958	0.500	100.0%	33.3%	0.0%
SigLIP2	Visual tune	0.976	0.986	0.800	100.0%	66.7%	0.0%

All stages are shown, including regressions. These are candidate results, not test-selected routes.

Unattended: zero-shot → probe → visual tune

Same held-out cohort: n=962; positives=242; negatives=720 • dev-max-F1 cutoffs

Model / source	Stage	AP ↑	ROC-AUC ↑	F1 ↑	Prec. ↑	Recall ↑	FPR ↓
MobileCLIP	Zero-shot	0.209	0.416	0.402	25.2%	100.0%	100.0%
MobileCLIP	Linear probe	0.361	0.667	0.404	25.3%	100.0%	99.0%
MobileCLIP	Visual tune	0.652	0.848	0.408	25.6%	100.0%	97.6%
SigLIP2	Zero-shot	0.309	0.675	0.090	9.3%	8.7%	28.5%
SigLIP2	Linear probe	0.855	0.911	0.402	25.2%	100.0%	100.0%
SigLIP2	Visual tune	0.896	0.933	0.403	25.2%	100.0%	99.6%

All stages are shown, including regressions. These are candidate results, not test-selected routes.

Smoking / Stanford40 photos: zero-shot → probe → visual tune

Same held-out cohort: n=806; positives=140; negatives=666 • dev-max-F1 cutoffs

Model / source	Stage	AP ↑	ROC-AUC ↑	F1 ↑	Prec. ↑	Recall ↑	FPR ↓
MobileCLIP	Zero-shot	0.220	0.578	0.297	17.4%	100.0%	99.7%
MobileCLIP	Linear probe	0.851	0.960	0.587	42.1%	97.1%	28.1%
MobileCLIP	Visual tune	0.901	0.972	0.678	52.3%	96.4%	18.5%
SigLIP2	Zero-shot	0.499	0.794	0.390	25.0%	88.6%	55.9%
SigLIP2	Linear probe	0.928	0.977	0.617	45.1%	97.9%	25.1%
SigLIP2	Visual tune	0.954	0.988	0.777	64.8%	97.1%	11.1%

All stages are shown, including regressions. These are candidate results, not test-selected routes.

The chosen route is not always the most fine-tuned model

Selection uses development mean source AP per target; simpler methods win ties

Target	MobileCLIP route	SigLIP2 route	Mobile AP	SigLIP AP
Smoking / CCTV	Visual tune	Visual tune	0.564	0.650
Littering / IWDD	Visual tune	Linear probe	0.858	0.885
Loitering	Visual tune	Zero-shot	0.093	0.097
Public sleep	Visual tune	Linear probe	0.976	0.915
Unattended	Visual tune	Visual tune	0.652	0.896

> **SigLIP keeps the probe for dumping/sleep and zero-shot for loitering, even where a test alternative scores higher.**

Unattended items: the cutoff changes the conclusion

Primary development-selected routes • same 962 endpoints, 242 positive / 720 normal

Model / source	Stage	AP ↑	ROC-AUC ↑	F1 ↑	Prec. ↑	Recall ↑	FPR ↓
MobileCLIP	Raw zero	0.652	0.848	0.408	100.0%	25.6%	0.0%
MobileCLIP	Dev max-F1	0.652	0.848	0.408	25.6%	100.0%	97.6%
MobileCLIP	Dev ≤5% FPR	0.652	0.848	0.631	47.0%	96.3%	36.5%
SigLIP2	Raw zero	0.896	0.933	0.749	96.7%	61.2%	0.7%
SigLIP2	Dev max-F1	0.896	0.933	0.403	25.2%	100.0%	99.6%
SigLIP2	Dev ≤5% FPR	0.896	0.933	0.403	25.3%	100.0%	99.4%

SigLIP raw F1 ≈0.749 and FPR ≈0.7%; its development-selected cutoffs produce ≈99% test FPR.

Keep seed 42 primary and report the other seeds

Each seed has its development-selected target route • same test sources in every repetition

Target	Mobile primary	Mobile range	SigLIP primary	SigLIP range
Smoking / CCTV	0.564	0.564–0.594	0.650	0.650–0.692
Littering / IWDD	0.858	0.799–0.861	0.885	0.885–0.885
Loitering	0.093	0.093–0.111	0.097	0.095–0.097
Public sleep	0.976	0.976–1.000	0.915	0.915–0.915
Unattended	0.652	0.513–0.697	0.896	0.855–0.902

These ranges measure optimization and route variation, not confidence intervals over unseen cameras.

After backbone training: regions, geometry and new heads

16 Oct expanded corpus • 330 configurations = 320 classifier fits + 10 text references

Candidate family	What changes	What stays controlled
Scene / predicted crop	Full scene plus contextual person/item crop	No source boxes select test crops
Ordered / geometry	Chronological views; sparse automatic box features	Timing and source splits retained
Weight interpolation	Pretrained ↔ earlier tuned visual weights	Alpha = 0, .25, .5, 1; selected on dev
Linear / MLP classifier	LBFGS probes or 64-unit ReLU / dropout .3	Same expanded training labels; source balancing
Selection	Per-target mean development-source AP	Primary seed 42; no best-test route replacement

> **Fitting a new classifier or blending saved weights is distinct from a new visual-backbone fine-tune.**

Five-target comparison on identical held-out cases

16 Oct CLIP strategies vs 19 Oct SmolVLM routed • same expanded train/dev and held-out labels

Cohort	n	Positive	Mobile AP	SigLIP AP	Smol AP
Smoking / CCTV	646	393	0.690	0.722	0.623
Littering / IWDD	100	50	0.806	0.893	0.766
Loitering	75	11	0.489	0.550	0.182
Public sleep	18	6	0.931	0.883	0.788
Unattended	962	242	0.980	0.925	0.693

> **This is a pipeline comparison: encoders, crops, resolution and adaptation methods differ.**

Photo and action-transfer comparison on matched cases

16 Oct CLIP strategies vs 19 Oct SmolVLM routed • same expanded train/dev and held-out labels

Cohort	n	Positive	Mobile AP	SigLIP AP	Smol AP
Smoking / Kaggle photos	224	112	0.997	1.000	0.984
Smoking / Stanford40	806	140	0.880	0.937	0.890
Smoking / HMDB action	906	109	0.576	0.664	0.784
Sleep / new authors	28	17	0.952	0.933	0.890
MEVA abandon action	27	15	0.545	0.621	0.646

> **This is a pipeline comparison: encoders, crops, resolution and adaptation methods differ.**

Matched cohorts: ranking gains versus operating errors

16 Oct selected CLIP strategies / 19 Oct routed SmolVLM • development-budget cutoffs

Model / source	Stage	AP ↑	ROC-AUC ↑	F1 ↑	Prec. ↑	Recall ↑	FPR ↓
MobileCLIP	CCTV	0.690	0.596	0.228	73.6%	13.5%	7.5%
SigLIP2	CCTV	0.722	0.647	0.086	75.0%	4.6%	2.4%
SmolVLM	CCTV	0.623	0.502	0.035	100.0%	1.8%	0.0%
MobileCLIP	Unattended	0.980	0.991	0.513	34.5%	100.0%	63.7%
SigLIP2	Unattended	0.925	0.965	0.403	25.3%	100.0%	99.4%
SmolVLM	Unattended	0.693	0.813	0.467	31.0%	94.6%	70.8%

High unattended AP still coexists with high normal FPR. The development 5% budget does not transfer.

LoRA adapts the answer function while vision stays frozen

SmolVLM-500M-Instruct • 3,973 training examples • five task-conditioned binary queries

Setting	Executed configuration
Updated weights	4,341,760 language-projection parameters; q/k/v/o and gate/up/down
LoRA setup	Rank 8; alpha 16; dropout 0.05; vision/connector/embeddings/output head frozen
Objective	Full-vocabulary first-answer-token CE for Yes / No; mask every input token
Optimizer	AdamW; LR 3e-5 or1e-4; weight decay .01; warmup 5%; gradient clip1.0
Effective batch	Microbatch 8 × accumulation 4 =32 examples; max 6 epochs; patience 2
Sampling / precision	Each train case once + equal balanced resample; BF16 train, float32 eval

No SmolVLM linear probe or full-parameter fine-tune was run; LoRA is the actual weight-adaptation stage.

All three adapter runs select epoch one

Primary seed 42 compares two learning rates; seed 43 repeats only the primary winning LR

Run / visual tower frozen	Epochs run	Best epoch	Best dev AP	Last train loss
s42 / 3e-05	3	1	0.658	0.319
s42 / 0.0001	3	1	0.703	0.223
s43 / 0.0001	3	1	0.655	0.215

More completed epochs do not imply a better selected model. Checkpoints maximize development macro source AP.

Original model → one joint LoRA → per-target routing

Same images, prompt and first-answer score within SmolVLM • primary seed 42

Cohort	n	Original AP	Joint AP	Routed AP	Routed weights
Smoking / CCTV	646	0.600	0.623	0.623	LoRA high LR
Littering / IWDD	100	0.592	0.766	0.766	LoRA high LR
Loitering	75	0.425	0.182	0.182	LoRA high LR
Public sleep	18	0.744	0.633	0.788	LoRA low LR
Unattended	962	0.476	0.693	0.693	LoRA high LR
Smoking / Stanford40	806	0.583	0.890	0.890	LoRA high LR

Routing improves the public-sleep choice; loitering ranking still falls below the original model.

SmolVLM: full decision metrics after adaptation

Primary per-target routes • maximum-development-F1 cutoffs • all five target cohorts

Model / source	Stage	AP ↑	ROC-AUC ↑	F1 ↑	Prec. ↑	Recall ↑	FPR ↓
Smoking / CCTV	Routed LoRA	0.623	0.502	0.754	62.0%	96.2%	91.7%
Littering / IWDD	Routed LoRA	0.766	0.803	0.789	70.3%	90.0%	38.0%
Loitering	Routed LoRA	0.182	0.415	0.182	11.4%	45.5%	60.9%
Public sleep	Routed LoRA	0.788	0.889	0.444	66.7%	33.3%	8.3%
Unattended	Routed LoRA	0.693	0.813	0.459	29.8%	100.0%	79.3%
Stanford40	Routed LoRA	0.890	0.959	0.731	63.4%	86.4%	10.5%

Full-vocabulary Yes-minus-No margins support ranking; these are not calibrated event probabilities.

SmolVLM: high endpoint recall can mean many false positives

Primary routed model • compare three frozen operating points on the same cases

Model / source	Stage	AP ↑	ROC-AUC ↑	F1 ↑	Prec. ↑	Recall ↑	FPR ↓
Smoking / CCTV	Raw zero	0.623	0.502	0.000	0.0%	0.0%	0.0%
Smoking / CCTV	Dev max-F1	0.623	0.502	0.754	62.0%	96.2%	91.7%
Smoking / CCTV	Dev ≤5% FPR	0.623	0.502	0.035	100.0%	1.8%	0.0%
Unattended	Raw zero	0.693	0.813	0.537	45.8%	64.9%	25.8%
Unattended	Dev max-F1	0.693	0.813	0.459	29.8%	100.0%	79.3%
Unattended	Dev ≤5% FPR	0.693	0.813	0.467	31.0%	94.6%	70.8%

Unattended at the dev-budget cutoff: 229/242 positives found, but 510/720 normals flagged.

A second seed exposes unstable task gains

Same high-rate joint architecture and source cohort • primary route remains seed 42

Cohort	Original AP	Joint seed42	Joint seed43
Smoking / CCTV	0.600	0.623	0.629
Littering / IWDD	0.592	0.766	0.766
Loitering	0.425	0.182	0.131
Public sleep	0.744	0.633	0.917
Unattended	0.476	0.693	0.705

Loitering regresses in both joint seeds; public-sleep AP varies strongly on only six positive test photos.

VadCLIP learns a sequence model on frozen CLIP-B/16

Corrected offline supervised smoking study • 1,910 train / 312 dev / 1,056 test frames

Frozen-feature controls

Two fixed text prompts
CLIP-B/16 image embeddings
Logistic probe C1, max 2,000 steps
Same training/test frames as adapter
Development-only F1 cutoffs

Actual adapter training

About12.642M trainable parameters
Two temporal layers + graph branches
Length 256; attention window 8
AdamW LR.001, decay.01; batch 4
Seed 234; max 100 epochs, patience 50

This is offline sequence-context evaluation. It does not establish a causal five-anomaly event detector.

VadCLIP: zero-shot → probe → temporal adapter

Same 1,056 held-out frames: 560 positive / 496 normal • per-stage dev-max-F1 cutoffs

Model / source	Stage	AP ↑	ROC-AUC ↑	F1 ↑	Prec. ↑	Recall ↑	FPR ↓
VadCLIP	CLIP text	0.482	0.410	0.693	53.0%	100.0%	100.0%
VadCLIP	Linear probe	0.544	0.487	0.685	52.9%	97.1%	97.8%
VadCLIP	Temporal adapter	0.633	0.587	0.565	62.1%	51.8%	35.7%
Reference	Always positive	0.530	0.500	0.693	53.0%	100.0%	100.0%

Adapter AP rises from0.482 to0.633, but FPR remains35.7% and recall51.8% at its frozen cutoff.

Four adapted models: metrics with their protocol labels

A: 646-frame camera holdout (393+) • B: 1,056-frame session holdout (560+) • dev-max-F1

Model / source	Stage	AP ↑	ROC-AUC ↑	F1 ↑	Prec. ↑	Recall ↑	FPR ↓
MobileCLIP • A	Visual tune	0.564	0.440	0.735	60.7%	93.1%	93.7%
SigLIP2 • A	Visual tune	0.650	0.531	0.727	60.3%	91.3%	93.3%
SmolVLM • A	Joint LoRA	0.623	0.502	0.754	62.0%	96.2%	91.7%
VadCLIP • B	Temporal adapter	0.633	0.587	0.565	62.1%	51.8%	35.7%

Metric columns are shared; supervision, training size and test split are not equal across all four rows.

Later learning adds automatic regions and temporal memory

16–17 Oct CLIP pilot • frozen pretrained visual encoders • 50 saved candidate heads

Component	Executed training / representation	Evaluation boundary
Input evidence	16 HMDB views; full / body / upper-body crops	CCTV smoking test remains static
Temporal classifier	128-unit projection; 64-unit causal GRU; optional18geometry channels	Past-only sequence features; no oracle tracks
Person supervision	1,460 matched train regions,277 positive; scene BCE + .5region BCE	Automatic test regions; no test box selection
Optimization	AdamW LR.0005, decay.01; batch 128; max 60 epochs; patience 8	Worst dev-source AP, then mean AP
Additional source data	530 train /161 dev /205 test HMDB clips after title-part purge	Different cohort from906-clip transfer test

> **These are new classifiers and temporal heads; the visual encoders are frozen in this pilot.**

Later smoking stages: CCTV frames, n646

Within-pilot training/control cohort • original pooled budget vs later every-source policy

Model / source	Stage	AP ↑	ROC-AUC ↑	F1 ↑	Prec. ↑	Recall ↑	FPR ↓
MobileCLIP	Scene probe	0.626	0.527	0.000	0.0%	0.0%	0.0%
MobileCLIP	Region / person	0.725	0.638	0.272	76.5%	16.5%	7.9%
MobileCLIP	Sourcewise policy	0.725	0.638	0.191	75.4%	10.9%	5.5%
SigLIP2	Scene probe	0.717	0.626	0.000	0.0%	0.0%	0.0%
SigLIP2	Region / person	0.688	0.629	0.044	69.2%	2.3%	1.6%
SigLIP2	Sourcewise policy	0.720	0.634	0.035	63.6%	1.8%	1.6%

Later policies change selection/calibration; the three rows do not isolate only a larger neural network.

Later smoking stages: Adapted HMDB test, n205

Within-pilot training/control cohort • original pooled budget vs later every-source policy

Model / source	Stage	AP ↑	ROC-AUC ↑	F1 ↑	Prec. ↑	Recall ↑	FPR ↓
MobileCLIP	Scene probe	0.842	0.959	0.685	58.1%	83.3%	10.3%
MobileCLIP	Region / person	0.908	0.979	0.852	83.9%	86.7%	2.9%
MobileCLIP	Sourcewise policy	0.908	0.979	0.867	86.7%	86.7%	2.3%
SigLIP2	Scene probe	0.888	0.972	0.750	70.6%	80.0%	5.7%
SigLIP2	Region / person	0.778	0.874	0.755	87.0%	66.7%	1.7%
SigLIP2	Sourcewise policy	0.964	0.994	0.821	88.5%	76.7%	1.7%

Later policies change selection/calibration; the three rows do not isolate only a larger neural network.

20 October: old-only, new-only and mixed training

740 new scripted-action clips • no new visual-backbone or SmolVLM weight training

Setting	Frozen experimental design
New labels	Smoking gesture vs drinking/eating/phone; masked actors; no cigarette confirmation
Source split	Train340/dev160/test240; scene blocks and numeric subject aliases disjoint
Classifier families	Old-only2,812/new-only340/mixed3,152 training examples
Candidate budget	2encoders ×3families ×2full/crop recipes ×3C values =36logistic fits
Selection and views	Same1,474 dev cases; worst-source recall at every-source≤5% FPR; 16newviews
Held-out scope	1,916cases per encoder:historical1,676 + new proxy240

More source data is tested as weak supervision. Scene4 has only three held-out subject groups.

Data-addition outcome: Historical CCTV

Frozen per-family development selection • every-source budget cutoff • same source cohort in all rows

Model / source	Stage	AP ↑	ROC-AUC ↑	F1 ↑	Prec. ↑	Recall ↑	FPR ↓
MobileCLIP	old only	0.694	0.599	0.179	74.1%	10.2%	5.5%
MobileCLIP	new only	0.610	0.512	0.000	0.0%	0.0%	0.0%
MobileCLIP	mixed	0.567	0.417	0.200	54.5%	12.2%	15.8%
SigLIP2	old only	0.731	0.655	0.160	77.8%	8.9%	4.0%
SigLIP2	new only	0.563	0.446	0.000	0.0%	0.0%	0.4%
SigLIP2	mixed	0.623	0.528	0.460	60.9%	36.9%	36.8%

SigLIP improves proxy AP but loses CCTV ranking and normal rejection; no new model is promoted.

Data-addition outcome: New action proxy

Frozen per-family development selection • every-source budget cutoff • same source cohort in all rows

Model / source	Stage	AP ↑	ROC-AUC ↑	F1 ↑	Prec. ↑	Recall ↑	FPR ↓
MobileCLIP	old only	0.498	0.783	0.000	0.0%	0.0%	0.0%
MobileCLIP	new only	0.455	0.829	0.000	0.0%	0.0%	0.0%
MobileCLIP	mixed	0.503	0.860	0.000	0.0%	0.0%	11.1%
SigLIP2	old only	0.357	0.697	0.000	0.0%	0.0%	0.0%
SigLIP2	new only	0.631	0.882	0.592	56.9%	61.7%	15.6%
SigLIP2	mixed	0.777	0.925	0.342	81.2%	21.7%	1.7%

SigLIP improves proxy AP but loses CCTV ranking and normal rejection; no new model is promoted.

New-source SmolVLM transfer after prior LoRA

No new training • original vs fixed prior smoking adapter •8 views each; 240 test clips,60 positive

Model / source	Stage	AP ↑	ROC-AUC ↑	F1 ↑	Prec. ↑	Recall ↑	FPR ↓
Smol original	New-source dev	0.385	0.709	0.454	35.9%	61.7%	36.7%
Smol original	19 Oct dev	0.385	0.709	0.000	0.0%	0.0%	0.0%
Smol lora	New-source dev	0.357	0.647	0.368	38.9%	35.0%	18.3%
Smol lora	19 Oct dev	0.357	0.647	0.000	0.0%	0.0%	0.0%

Original AP0.386 → LoRA0.357. Both older budget cutoffs miss all new-source positives.

What the four-model evidence supports for each target

Compare complete source/metric records; missing supervision cannot be filled with a score

Target	MobileCLIP / SigLIP2	SmolVLM	VadCLIP
Smoking	Photo gains; CCTV/source regressions persist	LoRA mixed CCTV gain; proxy transfer regresses	Temporal adapter AP gain; separate sessions
Littering	Visual/probe ranking improves on IWDD	LoRA ranking improves on IWDD	Not trained/evaluated
Loitering	Later geometry helps historical ranking	Both joint seeds regress	Not trained/evaluated
Public sleep	Strong small-photo diagnostic	Highly seed/route dependent	Not trained/evaluated
Unattended	High endpoint AP; threshold transfer poor	Recall gains with excessive FPR	Not trained/evaluated

> **No four-model winner across five real events is established by the current training studies.**

Detailed numbers and training choices are available to review

Editable slides + vector PDF + complete stage metrics + source identities

Slide companion files

Unrounded metrics and confusion counts
Per-stage cutoff / seed / source fields
Dataset counts with positive labels
Detailed speaker notes per slide
Source hashes and measured layout

What remains necessary

Compatible independent event labels
New camera/day source blocks
Drinking / eating / phone controls
Track duration and owner/item history
Frozen source-aware acceptance tests

Read stage comparisons alongside source coverage and FPR; more fine-tuning alone has not solved transfer.